

Analytical Problem Solving

OUTPUT

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IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help

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ARTIFICIAL INTELLIGENCE ASSESSMENT SOLVER
=====
1. Question 1: Water Jug Problem (BFS)
2. Question 2: Mars Rover PEAS & Architecture Analysis
3. Question 3: 8-Queens Problem (Backtracking)
4. Question 4: OLA Cab Booking Agent Simulation
5. Question 5: Uniform Cost Search (UCS) Network Solver
6. Exit
=====
Enter your choice (1-6): 1

--- Running Water Jug Problem Solver ---

Goal Reached! Total Steps: 6
Step 0: 4-Gal Jug = 0g, 3-Gal Jug = 0g
Step 1: 4-Gal Jug = 4g, 3-Gal Jug = 0g
Step 2: 4-Gal Jug = 1g, 3-Gal Jug = 3g
Step 3: 4-Gal Jug = 1g, 3-Gal Jug = 0g
Step 4: 4-Gal Jug = 0g, 3-Gal Jug = 1g
Step 5: 4-Gal Jug = 4g, 3-Gal Jug = 1g
Step 6: 4-Gal Jug = 2g, 3-Gal Jug = 3g

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Enter your choice (1-6): 2

--- Mars Rover PEAS & Architecture Analysis ---

[P] PERCEPTS:
- Cameras (images, depth maps), Infrared sensors, LiDAR range data.
- Temperature, pressure, wind gauges, wheel slippage trackers, spectrometers.

[E] ENVIRONMENT:
- Partially Observable, Stochastic, Sequential, Dynamic, Continuous, Single-agent.

[A] ACTIONS:
- Steering, driving, core drilling, soil scooping, data antenna adjustment.

[S] PERFORMANCE MEASURE:
- Scientific value of samples, system safety (not getting stuck), battery efficiency.

[ARCHITECTURE] RECOMMENDED:
- Goal-Based / Utility-Based Hybrid Agent.
- Reason: Earth-to-Mars latency is high. The rover must independently calculate the
  safest, highest-yield paths without waiting for immediate human reflex commands.

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--- Mars Rover PEAS & Architecture Analysis ---

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Enter your choice (1-6): 3

--- Running 8-Queens Problem Solver ---
Q . . . . .
. . . . . Q .
. . . . Q . .
. . . . . Q
. . . . . Q
. Q . . . . .
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--- Running 8-Queens Problem Solver ---
Q . . . . .
. . . . . Q .
. . . . . Q .
. . . . . Q .
. Q . . . . .
. . . Q . . .
. . . . . Q .
. . Q . . . .
. . . . . Q .
. . Q . . . .

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=====
Enter your choice (1-6): 4

--- OLA Cab Agent Simulation ---
Goal Request: (0, 0) -> (10, 10) | Tier: MINI
Matched: Driver_1 (mini)
Action Sequence: [Dispatch Cab] -> [Navigate via optimal traffic route] -> [Arrive]

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Enter your choice (1-6): 5

--- Running Uniform Cost Search (UCS) ---
Evaluating node 'S' | Cumulative path cost: 0
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Action Sequence: [Dispatch Cab] -> [Navigate via optimal traffic route] -> [Arrive]

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Enter your choice (1-6): 5

--- Running Uniform Cost Search (UCS) ---
Evaluating node 'S' | Cumulative path cost: 0
Evaluating node 'A' | Cumulative path cost: 1
Evaluating node 'C' | Cumulative path cost: 2
Evaluating node 'D' | Cumulative path cost: 3
Evaluating node 'B' | Cumulative path cost: 4
Evaluating node 'G' | Cumulative path cost: 4

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OPTIMAL PATH FOUND: S -> A -> C -> G
TOTAL MINIMUM COST: 4
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Enter your choice (1-6): 6

Exiting Program. Good luck with your assignment!
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